AWS Cloud Quest: Cloud Practitioner

Hello Everyone, I’ve recently completed “AWS Cloud Quest: Cloud Practitioner” on AWS Skill Builder. The course consists of 12 interactive training sessions, each centered around solving a specific business problem. For every session, AWS provides a guided solution and walks you through the implementation steps. At the end, you’re challenged to complete a hands-on task independently to reinforce your learning.
This approach not only strengthens your practical skills but also introduces you to a variety of AWS services and foundational cloud concepts.
Below, I’ve listed all the training modules along with their Business scenarios, Solutions, Technical Annotations and the AWS services covered in each.

Cloud Computing Essentials

Business Scenario:

The web team wants a solution to improve the reliability of their beach wave conditions page.

Solution:

Improve website reliability by migrating to Amazon S3 static website hosting.

Technical Annotations Steps:

1. Use a bucket policy to secure an S3 bucket
2. Enable static website hosting on the S3 bucket
3. Test access to the webpage hosted on the S3 bucket

Concepts Covered:

1. Amazon S3
2. Amazon S3 Access Management

Cloud First Steps

Business Scenario:

The island stabilization team wants to increase the reliability and availability of its current stabilization system.

Solution:

Deploy the stabilization system across two EC2 instances in separate Availability Zones.

Technical Annotation Steps:

1. Launch two EC2 instances in two AZs within the same Region.

Concepts Covered:

1. AWS Global Infrastructure Benefits
2. Amazon EC2
3. Amazon EBS
4. DNS overview

Computing Solutions

Business Scenario:

The school wants to upgrade its class scheduling system, currently running on an Amazon EC2 instance, to provide greater computing power and memory capacity.

Solution:

Scale up the Amazon EC2 instance by selecting a larger instance type to enhance the application's performance.

Technical Annotation Steps:

1. Change the EC2 instance to a larger instance type

Concepts Covered:

1. Amazon EC2
2. Amazon - System Manager

Network Concepts

Business Scenario:

The bank wants to establish a network architecture that securely controls communication between its internal resources and the internet.

Solution:

Configure VPC networking components, including routing tables, an internet gateway, and security groups, to enable secure internet connectivity.

Technical Annotation Steps:

1. Configure a routing table and attach an internet gateway.
2. Configure a security group.

Concepts Covered:

1. Amazon VPC overview
2. Amazon VPC Concepts (IP addressing(CIDR), Route tables)
3. Amazon VPC Security (NACLs, Security Groups)
4. Amazon VPC Internet Connectivity (Internet Gateway, NAT Gateway, Egress-only Internet gateway)

Cloud Economics

Business Scenario:

The surf shop wants a cost estimate for scalable cloud architecture to support their online store.

Solution:

Create a cost estimate for a varying Amazon EC2 architecture based on peak demand.

Technical Annotation steps:

1. Create an estimate using AWS Pricing Calculator
2. Calculate the cost of Amazon EC2 web servers
3. Save and share your estimate

Concepts Covered:

1. AWS Pricing
2. AWS Cloud Adoption Framework Overview

Databases in Practice

Business Scenario:

The insurance company wants to help its database administrators spend less time in operational tasks, such as patching and managing database infrastructure. They also want a solution that improves database availability and efficiency.

Solution:

Migrate to Amazon RDS to automate routine database administration tasks, while implementing Multi-AZ deployment and read replicas to provide high availability and improve performance.

Technical Annotation Steps:

1. Create an Amazon RDS DB instance.
2. Enable backups to your database.
3. Enable multiple AZs for your Amazon RDS Deployment.
4. Create an Amazon RDS read replica.

Concepts Covered:

1. Amazon RDS Overview
2. Amazon RDS - Lower admin Burden Performance
3. Amazon RDS - Availability & Durability
4. DNS Overview
5. Amazon RDS - Scalability

Connecting VPCs

Business Scenario:

The Marketing team wants to maintain separate network environments for each department while ensuring they can still communicate with one another.

Solution:

Implement VPC peering to enable communication between VPCs, allowing Marketing and Developer EC2 instances to access the Financial Services server in the Finance VPC.

Technical Annotation Steps:

1. Configure VPC peering between Marketing and Finance networks

Concepts Covered:

1. Amazon VPC overview
2. Amazon VPC Concepts
3. Amazon VPC Peering Connections

First NoSQL Database

Business Scenario:

The Streaming entertainment company wants to collect viewer behavior data in a high-performance, scalable database to enhance its streaming features.

Solution:

Implement Amazon DynamoDB to store and process viewer behavior data, including content consumption and device analytics.

Technical Annotation Steps:

1. Create an Amazon DynamoDB table.
2. Create a DynamoDB record with metadata attributes.

Concepts Covered:

1. NoSQL
2. Difference between SQL and NoSQL
3. How to create a NoSQL table
4. Amazon DynamoDB Overview
5. Amazon DynamoDB Queries Overview

File Systems in the Cloud

Business Scenario:

The pet modeling agency wants a way to share files across its branch offices without managing physical storage infrastructure.

Solution:

Implement Amazon EFS to provide a fully managed, scalable file system that can be accessed across multiple branch locations.

Technical Annotation Steps:

1. Set up Amazon EFS.
2. Create a mount target for the pet client photos repository.

Concepts Covered:

1. Amazon EFS Overview

2. Amazon EFS Benefits
3. Amazon EFS Features

Core Security Concepts

Business Scenario:

The stock exchange wants to restrict support engineer’s system access to only those actions required for their specific roles, enhancing overall security controls.

Solution:

Implement IAM groups with least-privilege permissions to control support engineers’ access to AWS resources.

Technical Annotation Steps:

1. Create an IAM group for support engineers.
2. Attach a managed EC2 read-only access policy to the IAM group.
3. Create and attach a user to the group.

Concepts Covered:

1. AWS Security and Compliance Overview
2. AWS IAM - Overview
3. AWS IAM - Manage Permissions
4. AWS IAM Features - Access Analysis

Auto-Healing and Scaling Applications

Business Scenario:

The gaming cafe wants to implement auto-healing servers and restrict users to a specific provisioning capacity.

Solution:

Configure scheduled scaling rules for an EC2 auto Scaling group

Technical Annotation Steps:

1. Create an Auto Scaling group with launch configurations.
2. Configure scheduled scaling.

Concepts Covered:

1. Auto Healing and Scaling
2. Amazon EC2 Overview
3. Amazon CloudWatch Overview

Highly Available Web Applications

Business Scenario:

The travel agency wants to make sure their web application remains available to customers even if some system components fail.

Solution:

Deploy the web application across multiple AZs with an Application Load Balancer and health checks to provide high availability.

Technical Annotation Steps:

1. Create an Auto Scaling group across multiple AZs
2. Create a load balancer.
3. Attach the Auto Scaling group to the load balancer.
4. Create a load balancer health check for EC2 instance

Concepts Covered:

1. Amazon Route53 Overview
2. Amazon CloudFront Overview
3. Amazon S3 Overview
4. Amazon ELB Overview
5. Amazon RDS Overview
6. Auto Healing and Scaling

This journey has deepened my understanding of AWS architecture and cloud problem-solving. I’m excited to apply these skills and continue sharing my learnings. Let me know if you’ve explored AWS Cloud Quest or have questions—I’d love to connect!